UTILITY-AWARE SOCIAL NETWORK ANONYMIZATION USING GENETIC ALGORITHMS

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INTRODUCTION

- Social networks can be modeled as a graph (nodes = people, edges = connections).
- Useful in tasks like robustness analysis [1], finding central nodes [2], and community detection.
- Often contains personally identifiable information → **privacy risk** [3, 4].

PROBLEM STATEMENT

How can network **anonymization** be formulated as an **optimization** problem to better balance anonymity and utility?

MEASURING ANONYMITY

- $G = (V, E)$ is undirected and unweighted.
- Nodes in V are connected by edges in E.
- $deg(v)$: number of connections for node v.
- $\Delta(v)$: nr. of edges between neighbors of v.
- Node v is unique if no another node shares the same $(deg(v), \Delta(v))$.
- $U(G)$: fraction of unique nodes in G.

ANONYMIZATION PROBLEM AND APPROACH

Given a graph G and budget Γ, delete a set of edges $E_{\mathbf{del}} \subset E$ with $|E_{\mathbf{del}}| \leq \Gamma$, maximizing anonymity $1 - U(G)$ while minimizing $|E_{\mathbf{del}}|$.

$B = (b_1, \ldots, b_{|E|})$, where $b_i = 1$ if edge i is deleted, else 0. G'_B is the graph with edges removed as indicated by B.

$$f(G'_B) \quad = \quad U(G'_B) + \text{penalty} \ \rightarrow \ \min$$

Example

Figure 1 provides an example of a small network with four unique nodes based on their $(\deg(v), \Delta(v))$ values shown in Table 1. The algorithm deletes edges 5 and 11, yielding the following bit string:

Edge	1	2	3	4	5	6	7	8	9	10	11	12	13
B	0	0	0	0	1	0	0	0	0	0	1	0	0

Comparing fitness values, Table 1 shows that G' (Figure 2) no longer contains any unique nodes.

Table 1. $(\deg(v), \Delta(v))$ *node states before and after anonymization; unique states in red.*

	1	2	3	4	5	6	7	8	9	10	f
G	(1,0)	(3,1)	(2,1)	(4,1)	(2,0)	(3,1)	(3,1)	(4,2)	(2,1)	(2,1)	**0.4**
G'_B	(1,0)	(3,1)	(2,1)	(3,1)	(2,0)	(2,1)	(3,1)	(3,1)	(1,0)	(2,0)	**0**

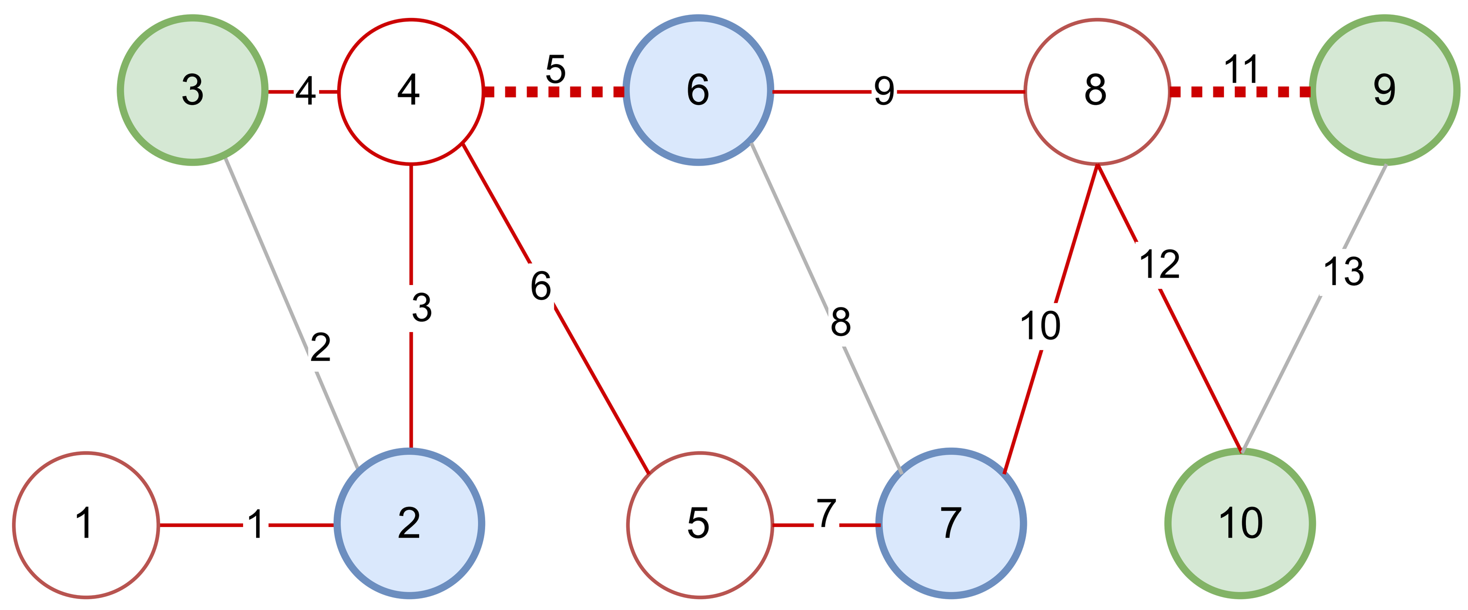

Figure 1: Graph G with four unique nodes

↓ **Anonymization**
(Edges 5 and 11 deleted)

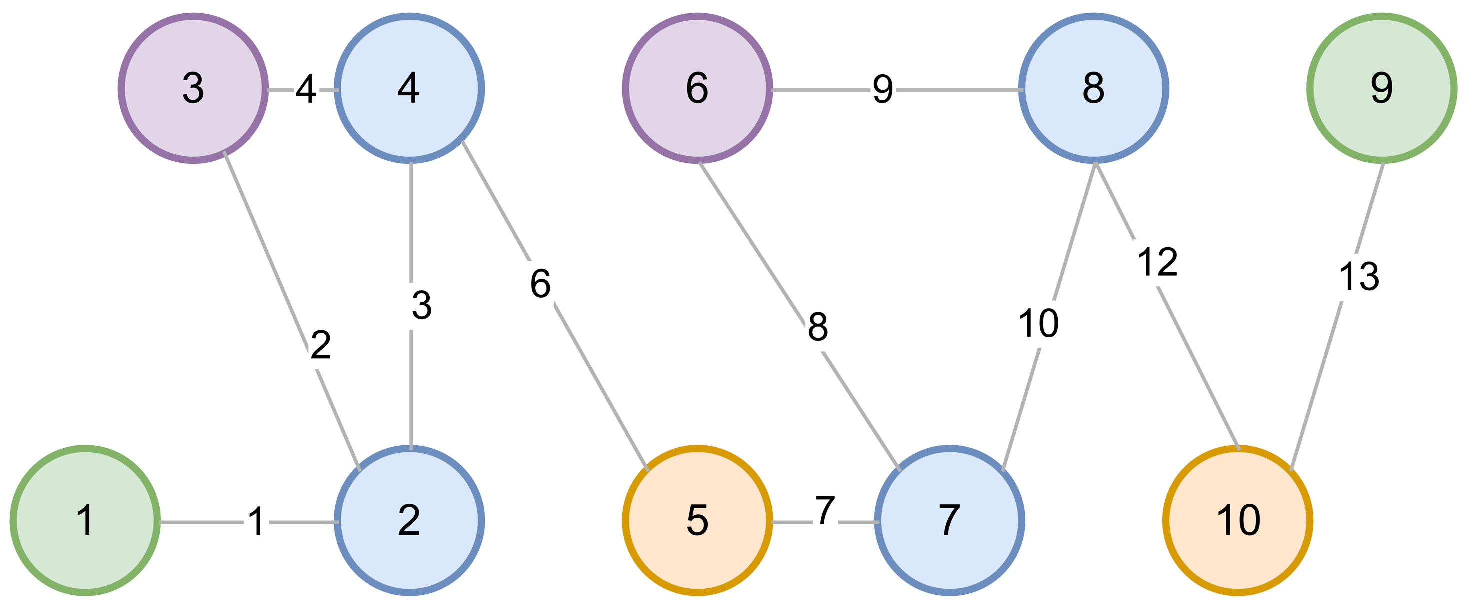

Figure 2: Anonymized graph, G'

ALGORITHMS

Genetic Algorithm (GA)
Using hyperparameter tuning:

- **Population**: $\mu = 100$, bits set at $p = 0.005$.
- **Selection**: roulette wheel; offspring $\lambda = 150$ via uniform or $c = 25$ crossover.
- **Mutation**: rate $\alpha = 0.05$, with decay.
- **Survivor**: $(\mu + \lambda)$-selection.

Uniqueness-aware GA (UGA)
We constrain edge selection to those connected to unique nodes (red edges in Fig. 1).

Baselines: ES samples edges at random [3]; UA maximizes unique affected nodes [4].

MAIN-TAKEAWAYS

Compared to baseline our GAs:

- Anonymize 14× more nodes than UA.
- Better preserve data utility.
- Match random sampling in downstream task performance.
- Offer strong anonymization with minimal utility loss.

RESULTS

Table 2. Performance. *Statistics of social network datasets, including number of nodes $|V|$, edges $|E|$, and proportion of unique nodes $U(G)$. Columns 6 to 9 show uniqueness reduction (higher is better); columns 10 to 14 report improvement factors between algorithms (higher is better). Results marked with † are not significant.*

	$\|V\|$	$\|E\|$	$U(G)$	$\|V\| \times U(G)$	ES	UA	GA	UGA	ES vs. GA	ES vs. UGA	UA vs. GA	UA vs. UGA	GA vs. UGA
FB Reed98	962	18,812	0.78	748	34	38	**391**	**391**	11.44	11.42	10.43	10.41	1.00 †
Blogs	1,224	16,715	0.49	598	11	13	**313**	310	29.79	29.50	23.23	23.01	0.90 †
FB Simmons	1,518	32,988	0.79	1,192	26	41	**585**	567	22.35	21.65	14.15	13.71	0.97 †
College msg.	1,899	13,838	0.24	454	31	38	308	**318**	10.10	10.43	8.11	8.38	1.03
GRQC collab.	5,242	14,484	0.05	285	2	0	192	**204**	100.84	107.16	-	-	1.06 †

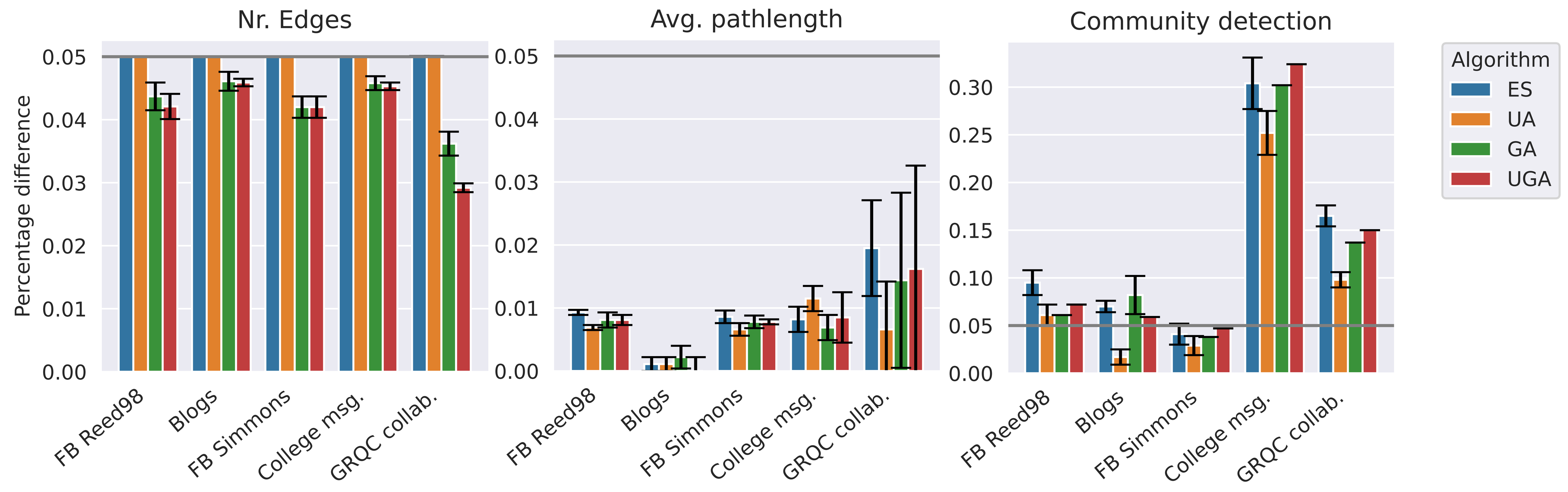

Figure 3: Utility. *Change in utility measures ± one standard deviation for each network, using the four algorithms (color).*

REFERENCES

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